1

Introduction

> "On the one hand, the perhaps evolutionary need to put everything into categories: predator, twilight, edible, [Reinforcement Learning, Causal Inference]. On the other hand, the need to tap into the great soup of being, in which we actually live."
>
> Maggie Nelson, [with edits from MaryLena Bleile]

1.1 Motivation

According to Richard Feynman, "one of the most dramatic moments in science is when two great fields suddenly come together and are unified" [49]. Causal Inference and Reinforcement Learning are certainly two great fields: Both have led to groundbreaking inventions such as personalized medicine and self-driving cars. "Optimal Control Using Causal Agents" shows the deep synergy between these two revolutionary fields which has emerged over the past several decades.

Causal Inference and Reinforcement Learning are teleologically aligned; both address the problem of optimal decision-making under uncertainty. Their methods, too, share deep mathematical similarity. For example, inverse probability of treatment weighting in marginal structural models employs the same ratio structure as importance sampling in off-policy evaluation, and the Expectation-Maximization algorithm can be reformulated as a myopic version of value iteration, i.e. value iteration with discount factor $\gamma = 0$ (see section 5.2 for details). Also, single timepoint snapshot of the advantage function $A_\pi(x,a) = Q_\pi(x,a) - V_\pi(x)$ in Reinforcement Learning can be mathematically identical to the blip function $\psi(a,x;\beta)$ in Causal Inference as long as one centers the mean of the blip function to ensure $E[\psi(A,x;\beta)|x] = 0$, and as long as the state and outcome spaces match. Despite this deep synergy between Reinforcement Learning and Causal Inference, however, the language and notation surrounding each approach is very different: Reinforcement Learning evolved from Computer Science, while Causal Inference emerged independently from statistics. Moreover, the fields of Causal Inference and Reinforcement Learning have very different standards and conventions when it comes to validation and publication of results.

Language gaps like the Causal/Reinforcement Learning divide are unfortunately common in science. Consider, for example, the logistic regression: The statistician learns this model in the context of maximum likelihood estimation under a Bernoulli model, and is taught to focus on the estimation properties such as parameter standard error. Meanwhile, the machine learning practitioner learns the same logistic regression model as a classifier trained by minimizing the cross-entropy loss. Note that the fitted coefficients are identical in both cases: Minimizing cross-entropy is the same operation as maximizing the Bernoulli log-likelihood, up to a sign. But the two communities have built different vocabularies and conventions around the same object; the result is that the expert fluent in one tradition may not immediately recognize what the other is doing. The Causal Inference and Reinforcement Learning split is the same phenomenon at a larger scale.

The dual nature of logistic regression is generally well-known in both the Statistical and the Machine Learning community. However, the intimate relationship between Causal Inference and Reinforcement Learning is *not* widely known; while Reinforcement Learning and Causal Inference can complement one another, many researchers remain entrenched exclusively in one camp or the other. In 2023 on PubMed, for example, 3939 articles mentioned Causal Inference but NOT Reinforcement Learning, 3700 articles mentioned Reinforcement Learning but NOT Causal Inference, while only 194 articles mentioned "Causal Reinforcement Learning", or both "Causal Inference" as well as "Reinforcement Learning"[1]. This unfortunate dichotomy drives a wedge into the middle of the densely connected web of research that gave rise to both fields, unnecessarily partitioning the knowledge graph into two smaller - and thus less powerful - components (see Figure 1.1).

The stark exclusivity of these encampments is a manifestation of a common cognitive trap, which I refer to here as *familiarity bias*. Familiarity bias is the human tendency to cling to known tools, regardless of whether those tools are useful or correct for the problem at hand. For example, several studies have shown that the safest time to have a heart attack is actually during one of the major cardiology conferences, when all of the cardiologists are away [73, 72, 127]. One reason for this, discussed at length in [45], is that cardiologists exhibit familiarity bias towards their preferred tool, heart stents, which they tend to apply lavishly, even if the patient's situation does not match the context in which heart stents are useful [160, 97]. In a more literal definition of the word "tool", many firefighters die from forest fires because they cannot drop their tools, subconsciously choosing to die beside their saw rather than to leave their heavy equipment behind and run [180, 135, 179].

It is difficult, if not impossible, to remove this cognitive bias. The problem can be mitigated, however, by expanding one's familiar toolset: The more - and the more diverse - tools in one's toolbox, the wider the breadth of problems which are solvable using those tools. One goal of this book is to put the tools of Causal Inference into the same toolbox as those of Reinforcement Learning: This unification of theory will allow the practitioner of Reinforcement Learning to view Causal Inference as a familiar tool (and vice versa) without feeling the need to learn the foundations of Computer Science or Statistics from the ground up to get there. The first four chapters of this book are oriented towards the first goal, providing an applications-oriented translation manual which focuses on the cross-talk and connections between the two fields.

The second goal of this book is to illustrate the translational landscape surrounding modern hybrid methods which leverage both causality and Reinforcement Learning. In addition to building bridges *within* Causal Reinforcement Learning, I wish to build bridges *between* Causal Reinforcement Learning and other fields, such as information theory, physics, and philosophy. These translational elements are critical to the modern decision-maker, especially to those who work in industry or academics who collaborate on large diverse teams of researchers. These external connections are most prominent in Chapters 6-7, wherein I discuss the translational positioning of various modern hybrid methods which are known as "Causal Reinforcement Learning". However, external connections are also present in the

[1] Search was performed using the advanced search capabilities on PubMed, allowing keyword to appear in any field of the article (abstract, body, etc.).

earlier chapters, and a discussion of philosophical implications and analogies is presented throughout the work.

1.2 Structure

"Optimal Control Using Causal Agents" (OCUsCA) is a translation manual for practitioners who know either Causal Inference or Reinforcement Learning and need to understand the other. Readers seeking a Causal Inference implementation guide should refer to [111], or [62] for a classic theoretical exposition. Readers seeking a Reinforcement Learning implementation guide should refer to the classic text by Sutton and Barto [164], or to [185] for a more recent reference. For a comprehensive theoretical treatment with proofs, see Bareinboim's recent textbook [11].

The first three chapters of OCUsCA contain some material which might be familiar to readers as undergraduate-level material. The purpose of these foundations is to facilitate discussion of the conceptual bridges between known concepts in Causal Inference and Reinforcement Learning, and to allow precise translation of notational conventions between the fields. Additionally, the Reinforcement Learning expert may be unfamiliar with statistical fundamentals. Similarly, the professional statistician might be unfamiliar with basic Dynamic Programming concepts and notation, which form the backbone of Reinforcement Learning. Statisticians may skip Part I, though RL Practitioners are recommended to read Part II even if they are already familiar with the methods therein; Part II is translational rather than expository, drawing connections and bridges back to the material in Part I.

I have attempted to balance the content between Reinforcement Learning and Causal Inference. However, some asymmetry is inevitable, since there is more material in the field of Reinforcement Learning, both in terms of breadth as well as mathematical depth. This is apparent in the fact that the introductory textbook for Reinforcement Learning [164] is much longer than the introductory textbook for Causal Inference [62]. The discussion of Reinforcement Learning methods also typically requires more mathematical background. My approach to this issue is to keep the discussion Reinforcement Learning concepts generally broader with less technical detail. I have also kept the focus on value-based methods, which I believe are somewhat more interconnected to Causal Inference than other methods.

A fully comprehensive academic textbook which builds the field of Causal Reinforcement Learning from the ground-up is in production by Elias Bareinboim, cited previously. OCUsCA differs in several key regards. OCUsCA is not a comprehensive academic textbook - indeed, there are many important methods which are not included in this book - rather it is a translational monograph directed at practitioners already familiar with Causal Inference and Reinforcement Learning. In OCUsCA, I do not purport to provide an in-depth, comprehensive introduction to any field, since these already exist. Instead, I wish to highlight the position of the reader's field (Causal Inference or Reinforcement Learning) as a point in the densely connected web of science. Figure 1.2 highlights this idea: Bareinboim builds the mountain; OCUsCA builds the bridges.

Naturally, OCUsCA is written for the current practitioner of either Reinforcement Learning or Causal Inference. I will assume knowledge of basic probability (including expected values, Bayes' theorem, and conditional probabilities), calculus, and linear algebra throughout the text. Some proof sketches are included in the appendices; these include some functional analysis. However, the functional analysis is kept at a conceptual level, and major theoretical concepts such as the Banach Fixed-Point Theorem are not discussed in detail.

1.3 Title

The meaning of the word "agent" has rapidly evolved since the time of OCUsCA's conception (2024) to its completion (2026). An "agent", as the term is used here in the title, refers to any decision-making system which can learn from data. OCUsCA is not about Large Language Models (LLMs), though many LLMs may be foundational on the concepts herein. As the term grew in popularity under different connotations, we considered changing the name to avoid ambiguity, but since OCUsCA has already received some attention and recognition under its current name, we ultimately opted to keep it as it is.

1.4 Programming

Before turning to the substantive content, we should briefly address the programming conventions used throughout the book. The divide between Causal Inference and Reinforcement Learning is clearly apparent in the difference between their preferred programming languages of choice: `R` for Causal Inference and `Python` for Reinforcement Learning. We will see in later chapters that the nuances which distinguish these programming languages from one another reflect the different styles of thinking between Causal Inference and Reinforcement Learning. In this book we use both languages: Causal Inference examples are mostly written in `R` while Reinforcement Learning examples are mostly written in `Python`. We assume that the reader is familiar with at least one of these languages; the Causal Inference practitioner likely works in `R`, and the Reinforcement Learning expert indubitably knows `Python`.

R is an open-source scripting language which evolved from a language called *S*. Open-source languages are largely community-built; all of the development code for these types of languages is freely available online. As a result, anyone can easily propose contributed updates or packages - this is an attractive feature for research transparency, but it also means that support for established packages is not guaranteed. Some industries prefer to use the language SAS for this reason; SAS is proprietary (closed-source) software. The technical support for SAS is very reliable, but the caveat is that it is expensive. Moreover, not all modern statistical methods have convenient SAS implementations; researchers will typically create packages in R to wrap their methods. For this reason, many industries are adopting R despite its caveats.

Bayesian statisticians, in particular, tend to use R more than SAS since Bayesian analyses typically require more complicated programming and customization for each analysis. There is also a strong degree of overlap between the Bayesian and the Causal communities of statisticians. Two specific reasons account for this: i) the potential-outcomes framework ([136]) casts Causal Inference as a missing-data problem, and Bayesian inference extends naturally to causal effects framed this way (see [138]); and ii) Bayesian hierarchical models naturally accommodate heterogeneous treatment effects across subgroups, which is central to the causal questions we examine in Chapters 2 and 3. We will use R for most pure Causal Inference code examples.

Reinforcement Learning applications are usually written in the programming language Python. Python is also an open-source scripting language, but unlike R, Python is used by a wide variety of computer scientists and developers in addition to data scientists and statisticians. A full-stack engineer, for example, might use Python for some data-related procedures, but it is very unlikely that she would use R for any task: Python is more general than R. As a result, in part, of this generality, Python allows for more flexible implementation of Neural Network architectures (see Chapter 5). Since Reinforcement Learning evolved from Computer Science, it is therefore unsurprising that Reinforcement Learning practitioners prefer Python. Most of the code examples involving Reinforcement Learning concepts are therefore written in Python.

The syntactic differences between R and Python reflect deeper philosophical distinctions between Causal Inference and Reinforcement Learning; the chosen language of each field reflects how its practitioners conceptualize sequential decision problems. As we progress through the mathematical frameworks, we will see how R's functional, vector-oriented design naturally accommodates the marginalization operations of Causal Inference, while Python's object-oriented architecture provides elegant support for the agent-environment dynamics of Reinforcement Learning. A summary of key differences between R and Python is available in Table 1.1; more detailed explanations for each are available in appendices A and B.

R	**Python**
1-indexed (first element at position 1)	0-indexed (first element at position 0)
Vector-oriented: operations automatically vectorized	Scalar-oriented: explicit loops unless using NumPy
Pass-by-value with copy-on-write semantics	Passes object references by value (mutable objects can be modified in-place)
NA is single missing data type that propagates predictably	Multiple missing types (NaN, None, pd.NA) with different behaviors
Vector recycling automatic (shorter vectors repeat)	No automatic recycling; requires explicit broadcasting
Native data.frame with flexible column access	DataFrame requires pandas; stricter copy/view semantics
Functional paradigm with global workspace	Object-oriented with encapsulated state

TABLE 1.1: Critical Differences Between R and Python for Causal RL Implementation

1.5 Notation

Part of the difficulty in transferring knowledge of Reinforcement Learning to Causal Inference (and vice versa) is the differing terminology and notation between the two fields. OCUsCA introduces them both in a unified notational schema. Naturally, this unified notational schema differs from some conventional practices in each specific field.

> When I change the notational schema from convention in either field, I will indicate this by also providing the conventional notation in a shade box.

OCUsCA will follow the statistical convention of denoting random variables with capital letters, e.g. A, R, X. Specific instances of these variables will always be denoted using small letters, e.g. $P(A = a) = 0.3$. I will always use j to denote the index of a Markov Decision Process. If the Markov process is episodic (such as in the case of a clinical trial with time-varying covariates), then episodes (subjects) will be indexed using i and timepoints within episodes will be indexed using t: j then ranges over all (i, t) pairs, taking one unique value per subject-timepoint combination. If there is only one observation per subject, then $j = i$.

Table 1.2 gives a master table of notation. Note that this is a master reference for translating between Causal Inference and Reinforcement Learning notation. It is not meant to be read linearly on first encounter; most of the entries will only become meaningful as you progress through the book. Table 1.2 is a Rosetta Stone you keep open alongside the text.

TABLE 1.2: Master Notation Reference

Concept	**This Book**	**RL**	**CI**	**Notes**
Basic Elements				
State (random var.)	X	S	X	CI uses specific names
Specific state	x	s	x	
Observed state	X_o	o	X_{obs}	
Missing state	X_m	h or s_{hidden}	X_{mis}	
State space	Ω	$\mathcal{S}$	$\mathcal{X}$	
Action/Treatment	A	A	T or A	
Specific action	a	a	t or a	
Action space	$\mathcal{A}$	$\mathcal{A}$	$\{0, 1\}$	Binary in CI
Action k	$a^{(k)}$	a_k	$a = k$	
Reward/Outcome	R	R	Y	
Specific reward	r	r	y	
Reward function	$\rho(x, a)$	$r(s, a)$	-	
Potential Outcomes and Causality				
Potential outcome	$R\|A = a$ or $r\|a$	$E[G_j\|X_j = x, A_j = a]$	Y^a or $Y(a)$	Equal under exchangeability; see Ch. 2
Counterfactual	R_{cf}	$\mathbb{E}[G_j\|X_j = x, A_j \neq a]$	$\mathbb{E}[Y(1 - A)]$	Alternative policy outcome
Treatment effect	$\delta = r\|a^{(2)} - r\|a^{(1)}$	$Q(x, a) - Q(x, a')$	τ	Pairwise comparison
Blip/Advantage	$\psi(a, x; \beta)$	$A^\pi(x, a) = Q - V$	$\psi(a, x; \beta)$	Mean-centered effect
ATE	$E[R\|a^{(2)}] - E[R\|a^{(1)}]$	$E[Q(x, a) - Q(x, a')]$	$E[Y(1) - Y(0)]$	Average treatment effect
CATE	$E[R\|a^{(2)}, X] - E[R\|a^{(1)}, X]$	$Q(x, a) - Q(x, a')$	$\tau(x)$	Conditional ATE
Causal risk diff.	$r\|a^{(2)} - r\|a^{(1)}$	-	$Y(1) - Y(0)$	Individual level
G-formula	$E[R\|\pi(A = a)]$	$V^\pi(x)$	$E[Y\|\text{do}(a)]$	Interventional expectation
Do-operator	$P(R\|\pi(a\|x) = 1)$	$P(R\|\pi(a\|x) = 1)$	$P(Y\|\text{do}(x))$	Intervention distribution
Off-policy eval	$E_{\pi_t}[R]$ via $\hat{\pi}$	IS/PDIS	IPW estimation	Target from behavior
Structural eqn	$X' = f(X, A, \epsilon)$	$X' \sim \Gamma(\cdot\|X, A)$	$Y = f(X, U)$	Generative model
Sequential weights	$\prod_{t=1}^{j} w_{jt}$	$\prod_{t=1}^{j} \rho_t$	$\prod_{t=1}^{j} sw_{it}$	Cumulative importance
MSM weight	$w_j = 1/\hat{\pi}(a_j\|x_j)$	ρ_j	w_{ij}	Time-varying confounding
Policies and Value Functions				
Policy	$\pi(a\|x)$	$\pi(a\|s)$	$P(A = 1\|X)$	Treatment mechanism
Behavior policy	π_b	π_b or μ	$g(a\|x)$	Data generating policy
Target policy	π_t	π	$d^{opt}(x)$	Policy evaluated

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TABLE 1.2 – continued from previous page

Concept	This Book	RL	CI	Notes
Optimal policy	π^*	π^*	$d^*(x)$	
Value function	$V^\pi(x)$	$V^\pi(s)$	-	Expected return
Optimal value	$V^*(x)$	$V^*(s)$	-	
Q-function	$Q^\pi(x,a)$	$Q^\pi(s,a)$	$E[Y\|X,A]$	Action value
Optimal Q	$Q^*(x,a)$	$Q^*(s,a)$	-	
Advantage	$A^\pi(x,a)$	$A^\pi(s,a)$	$\psi(a,x)$	$Q^\pi - V^\pi$
Process Notation				
Trial/iteration	j	t	-	MDP iteration
Subject/episode	i	e or i	i	
Time in episode	t	t	t	
Treatment arm	k	-	k	
Combined index	j	-	-	One value per (i,t) pair
State at j	x_j	s_t	X_i	
Action at j	a_j	a_t	A_i	
Reward at j	r_j	r_t	Y_i	
Next state	x_{j+1}	s_{t+1}	$X_{i,t+1}$	
State history	$\bar{X}_j$	h_t	$\bar{X}_t$	States up to j
Action history	$\bar{A}_j$	$\bar{a}_t$	$\bar{A}_t$	Actions up to j
MDP Components				
Transition dynamics	$\Gamma(x'\|x,a)$	$P(s'\|s,a)$	$P(X'\|X,A)$	State transition
Discount factor	γ	γ	-	In $[0,1)$; $\gamma=0$ is myopic
Return	G_j	G_t	-	Discounted sum
Bellman operator	$\mathcal{B}$	T	-	
Bellman optimality	$\mathcal{B}^*$	T^*	-	
Learning Parameters				
Learning rate	α	α	-	Step size
Exploration rate	ϵ	ϵ	-	For epsilon-greedy
TD parameter	λ	λ	-	For TD(lambda)
Eligibility trace	$e_j(x)$	$e_t(s)$	-	
Model parameters	θ	θ or w	β	
Estimated params	$\hat{\theta}$	$\hat{\theta}$	$\hat{\beta}$	
Neural net weights	θ	θ or W	-	Deep learning
GARCH params	$(\omega,\alpha_g,\beta_g)$	-	-	Volatility model
Missing Data and Confounding				
Missingness indicator	M	-	R or M	$Here, M_{jk} = 1$ if observed (though much of the missing-data literature uses $M=1$ for missing)
Propensity score (Behavior policy)	$\hat{\pi}(a\|x)$	$\pi^b(a\|s)$	$e(x)$	$P(A=1\|X)$
IPTW weight	w_j	ρ or w	w_i	Inverse prob weight
Stabilized weight	sw_j	-	sw_i	Ratio form
Importance ratio	ω_j	ρ_t	$w(a\|x)$	Policy ratio

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TABLE 1.2 – continued from previous page

Concept	**This Book**	**RL**	**CI**	**Notes**
Belief state	$b(x)$	$b(s)$	-	POMDP
Belief update	$b_{j+1}(x')$	$b'(s')$	-	Bayes update
Observation function	$P(x_o \vert x, a)$	$O(o \vert s', a)$	-	POMDP
Specialized Notation				
Indicator function	$\mathbb{1}(\cdot)$	$\mathbb{1}(\cdot)$	$I(\cdot)$	
Expectation	$E[\cdot]$	$\mathbb{E}[\cdot]$	$E[\cdot]$	
Variance	$\mathrm{Var}(\cdot)$	$\mathrm{Var}(\cdot)$	σ^2	
Probability	$P(\cdot)$	$P(\cdot)$	$P(\cdot)$	
Conditional prob	$P(A \vert B)$	$P(A \vert B)$	$P(A \vert B)$	
Intervention as policy	$\mathbb{1}(\pi(A=a)=1)$	$\mathbb{1}(\pi(A=a)=1)$	$P(Y \vert \mathrm{do}(x))$	Degenerate policy implementing an intervention
TD error	δ_j	δ_t	-	Temporal difference
Kalman gain	K_j	K_t	K_t	
State covariance	Σ_x	Σ	Σ	
Process noise	u_j	w_t	ϵ	u for "unmodeled noise"
Observation noise	ν_j	v_t	η	Pointy cursive ν for "visible noise"; distinguishes from u_j
DAG structure	$\mathcal{G}$	$\mathcal{N}$	$\mathcal{G}$	Causal graph

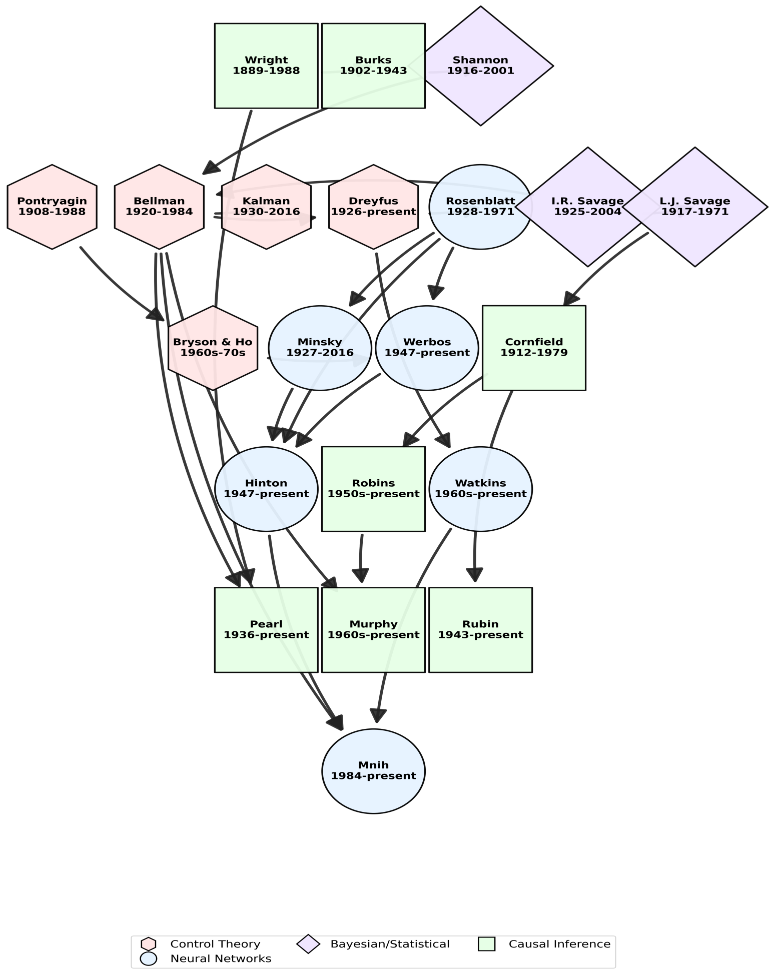

FIGURE 1.1: Network graph consisting of some inferred interactions (either social, or citations) between some of the prominent figures in the development of Causal Inference and Reinforcement Learning. Figure is the author's personal interpretation of the influence trajectory based on citations, co-authorship, and historical documents.

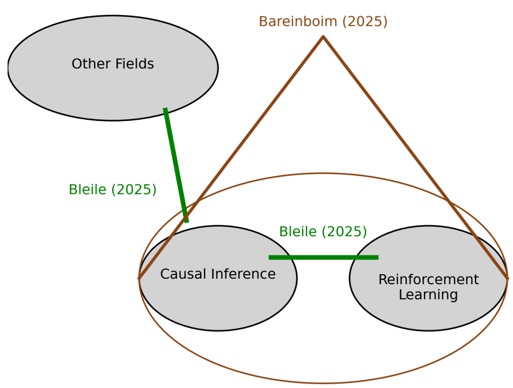

FIGURE 1.2: Illustration of the difference between this book and the textbook by Bareinboim (2025).